

Vectors

Exercise

Vectors

Take a look at the code below:

```
# Assign grades to objects
albert_grade <- 80
karla_grade <- 95
ariel_grade <- 90
geoff_grade <- 90

# Adjust grades by 5 points
albert_grade <- albert_grade + 5
karla_grade <- karla_grade + 5
albert_grade <- albert_grade + 5
geoff_grade <- geoff_grade + 5
```

What is the value of albert_grade after this code runs?

- a. 80
- b. 85
- c. 90
- d. 95

Exercise

Vectors

Take a look at the code below:

```
# Assign grades to objects
albert_grade <- 80
karla_grade <- 95
ariel_grade <- 90
geoff_grade <- 90

# Adjust grades by 5 points
albert_grade <- albert_grade + 5
karla_grade <- karla_grade + 5
albert_grade <- albert_grade + 5
geoff_grade <- geoff_grade + 5
```

What is the value of albert_grade after this code runs?

- a. 80
- b. 85
- c. 90
- d. 95

Imagine that we were dealing with grades from 50 students...

How likely would it be for us to make a mistake?

Vectors

Stores a sequence/array of values.

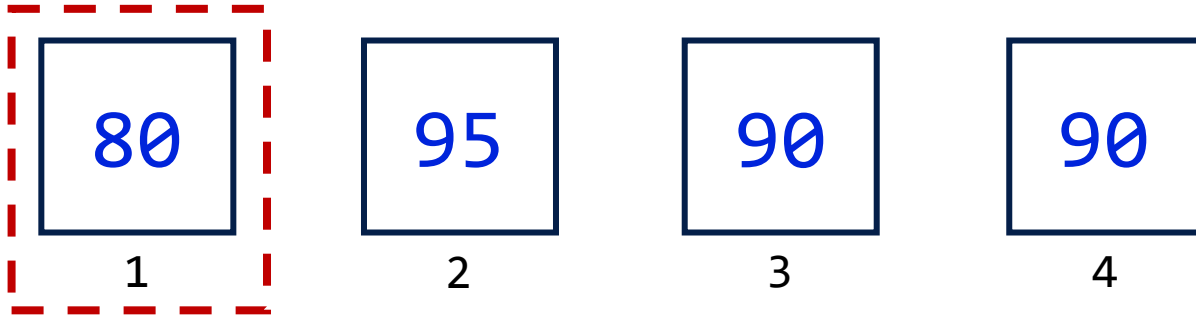
```
grades <- c(80, 95, 90, 90)
grades <- grades + 5
```

```
# Other type of vectors
numeric_vector <- c(1.1, 50.0, 92.562)
char_vector <- c("pizza", "soda pop", "3")
bool_vector <- c(TRUE, FALSE)
```

Vectors

Values can be **indexed** by the position

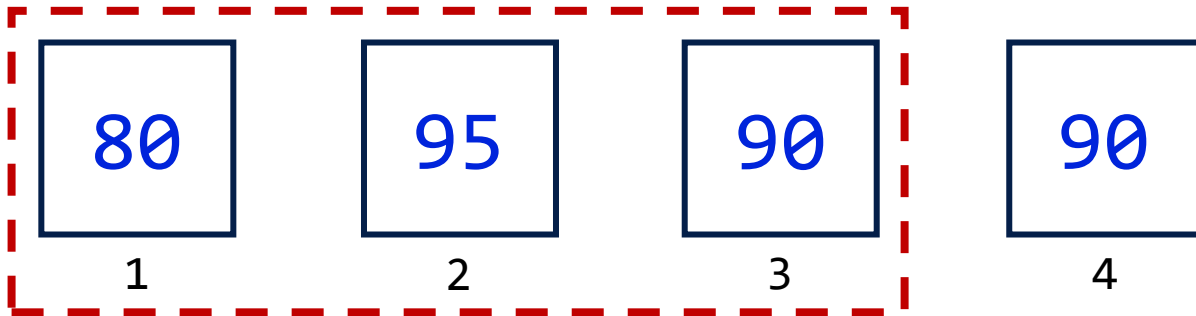
```
grades <- c(80, 95, 90, 90)  
grades[1] <- grades[1] + 5
```



Vectors

Values can be **indexed** by the position

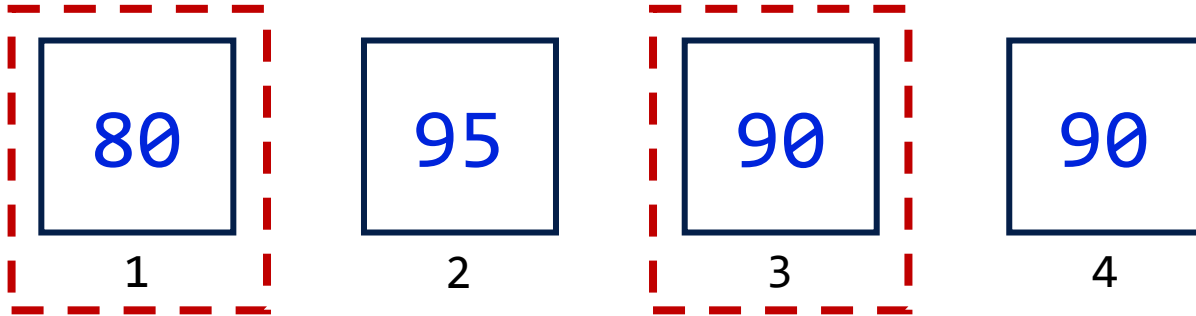
```
grades <- c(80, 95, 90, 90)  
grades[1:3] <- grades[1:3] + 5
```



Vectors

Values can be **indexed** by the position

```
grades <- c(80, 95, 90, 90)  
grades[c(1, 3)] <- grades[c(1, 3)] + 5
```



Vectors

Values can be **indexed** by logic

```
grades <- c(80, 95, 90, 90)  
grades[grades > 92] <- grades[c(1, 3)] + 5
```

80

FALSE

95

TRUE

90

FALSE

90

FALSE

Vectors

Values can be **indexed** by logic

```
grades <- c(80, 95, 90, 90)  
grades[c(FALSE, TRUE, FALSE, FALSE)]
```

80

FALSE

95

TRUE

90

FALSE

90

FALSE

Demo

Vectors

In this demo, we will:

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Knowledge Check

Vectors

Which of the following code snippets outputs the following:

[1] 4 7 3

a)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[1:4])
```

b)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[basic_vec >= 2])
```

c)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[c( 2, 3, 4 )])
```

Knowledge Check

Vectors

Which of the following code snippets outputs the following:

[1] 4 7 3

a)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[1:4])
```

b)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[basic_vec >= 2])
```

c)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[c( 2, 3, 4 )])
```



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